

MATÍAS SEPÚLVEDA

Software Engineer | Distributed Systems & Platform Engineering

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PROFESSIONAL SUMMARY

Software engineer with 5+ years of professional experience, including freelance development, progressing from full-stack applications to distributed backend systems and cloud-native platform engineering. Experience spans regional financial-market infrastructure and end-to-end ownership of production systems, from architecture and Terraform-managed infrastructure through implementation, deployment, and operation. Currently the sole technical owner of a 10-service platform supporting a 700-person operation, with a recent focus on Go, TypeScript, GCP, event-driven architecture, security, and AI-enabled workflows. Translates business requirements into pragmatic, maintainable systems while working effectively across established engineering teams and high-ownership environments.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, SQL | Spanish (Native), English (Advanced)

Cloud & Infrastructure: GCP (Cloud Run, Cloud SQL, Cloud Build, Pub/Sub, Cloud Tasks, GCS, Secret Manager, IAM, VPC, Artifact Registry), Terraform, Docker, CI/CD (GitHub Actions, Cloud Build)

Backend: Goa v3, GORM, Node.js, Express.js, REST APIs, gRPC, Webhooks, PostgreSQL, MySQL

Distributed Systems & Architecture: Service-Based Architecture, Distributed Services, Event-Driven Architecture, System Design, BFF/API Aggregation, Asynchronous Processing, Idempotency, Retry/Backoff, Infrastructure as Code, ADRs, Component/Deployment/Data-Flow Diagrams

Reliability & Security: Automated Testing, Health Endpoints, Standardized Error Handling, Audit Logging, JWT RS256, Hybrid RBAC, Cloud Run IAM, Service-to-Service Authentication, Secrets Management

AI: Vertex AI, Gemini, Qwen, RAG Context, Google Search Grounding, Goa-AI Agents, LLM APIs & Provider Fallbacks

Frontend & Additional: Next.js, React, React Native, Tailwind CSS, Radix UI, i18n (ES/ZH), PDFKit

Familiar / Learning: OpenTelemetry, Kubernetes, AWS

PROFESSIONAL EXPERIENCE

Software Engineer

Platform & Infrastructure | Sole Technical Owner

China First Highway Engineering Co. Ltd. Agencia en Chile (CFHEC)

Sep 2025 - Present | On-site | Chile

First engineering hire; established the company's software platform, infrastructure, and delivery practices from zero for a 700-person operation.

- Built and operate a cloud-native platform of 10 independently deployed Go services (Goa v3) and a Next.js frontend on GCP, giving a 700-person operation a single foundation for HR, recruitment, workflows, storage, inventory, and audit capabilities.
- Provisioned and standardized the platform through 11 modular Terraform modules covering Cloud Run, networking, IAM, Cloud SQL, Pub/Sub, Cloud Storage, Secret Manager, Cloud Build, and load balancing, enabling repeatable environments and infrastructure changes.
- Redesigned authentication around RSA-signed JWTs and server-side permission validation, reducing token payloads from approximately 2-4 KB to 200-300 bytes; documented the decision and six additional architectural trade-offs through ADRs.
- Designed an event-driven audit and resource-history system with Pub/Sub, plus a configurable approval engine using Cloud Tasks for timeout enforcement and retries, improving traceability and reliability for asynchronous workflows.
- Delivered a dedicated storage service with GCS integration, departmental quotas, secure streaming, inline previews, and mirror-based sharing, centralizing document handling without duplicating binary assets.
- Established quality and operational safeguards for critical Go services with in-memory SQLite tests, GitHub Actions, Cloud Build, and application/dependency health endpoints.
- Integrated BUK HR webhooks to synchronize personnel data for access-control and attendance workflows, and built a recruitment platform that has managed CV submissions from 1,500+ applicants.
- Developed inventory workflows for material reception, warehouse stock control, dispatch-guide documentation, low-stock alerts, and QR-based tool loans and returns tied to workers.
- Integrated applied AI into procurement and HR workflows using Vertex AI with Google Search grounding and Qwen LLM, enabling supplier-quotation comparison, bilingual reporting, and automated role generation.
- Partner with Chinese and Chilean leadership to translate business objectives into technical roadmaps, platform architecture, and delivery plans.
- Site Infrastructure Leadership: Led client-side delivery and provider coordination for dedicated 1 Gbps connectivity, seven network racks, 36 wireless access points, dark fiber, and a managed Fortinet FortiGate 90G firewall.

Software Engineer

Gatblac (Fixed-term contract supporting nuam)

Jun 2025 - Sep 2025 | Remote | Chile

- Built high-reliability backend microservices and PostgreSQL data-access layers in Go (Goa v3, GORM) to power cross-market securities-lending workflows connecting the Santiago, Lima, and Colombia stock exchanges.
- Partnered with cross-functional Product, QA, DevOps, and Architecture teams to deliver compliant financial infrastructure supporting multi-currency exchange operations.

Software Engineer

Valora Labs (Valora Empresas holding)

Mar 2024 - May 2025 | Remote | Chile

- Developed and modernized operational applications for industrial clients, including Minimiza for industrial-waste management and classification, Aprueba for internal courses and surveys, and Almacenamiento for facility storage management.
- Designed and deployed platform auth features, including a QR-based authentication flow with temporary, route-scoped dynamic RBAC to secure physical facility management.
- Maintained shared authentication, audit-history, permissions, and role-assignment services, applying ISO/IEC 27001-aligned security practices while supporting secure client operations.
- Owned features end-to-end across Node.js and Express.js APIs, React interfaces, PostgreSQL and MySQL schemas, production deployment, troubleshooting, and query optimization.

IT Department Lead / Full Stack Developer

Atalia

Aug 2021 - Mar 2023 | On-site | Chile

- Led a small technical function; defined development processes, handled vendor selection and IT acquisitions, and managed database design and server configuration.
- Led full-stack development of a real-estate management platform using React and Node.js from requirements gathering through production deployment.
- Built core modules for property listings, client lead tracking, document management, and business workflow digitization.

Freelance Development / Career Break

Mar 2023 - Mar 2024

SELECTED PROJECTS

AI Agent Observability & Regression Tracker (2026)

Creator & Developer | Python, FastAPI, PostgreSQL, Next.js, OpenAI API, Docker | github.com/Malygod

- Built an applied-AI engineering tool for ingesting agent traces, comparing performance, and identifying regressions across agent versions.
- Implemented SHA-256 input normalization and version comparison to surface regressions and performance deltas between agent versions.
- Integrated LLM-assisted diagnostics for exception traces and nested execution steps to identify probable failure causes and propose codebase fixes.

WorkoutSync — Product & Mobile Application Prototype

Creator & Developer | React Native, Product Design, Social & Activity Platform

- Conceived and developed a mobile platform to reduce barriers to physical activity by helping people discover activities, locations, and communities aligned with their interests and capabilities.
- Designed social and event-based interactions that allow users to organize and participate in local activities with others.
- Developed the concept from research into sedentary behavior, motivation, and social participation, emphasizing reduced friction between intention and real-world participation.

EDUCATION

Bachelor's Degree in Software Engineering (Ingeniería en Informática) - Completed

Universidad Tecnológica de Chile INACAP